



Mixed-Cation Validation Report

Editable local report test

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Pending researcher signature

KEYWORDS perovskite, mixed-cation, validation

01 | EXECUTIVE SNAPSHOT

Decision-ready project summary

PRJ-2026-014

The leading formulation remains FA0.90MA0.10.

RESEARCH OBJECTIVES

Compare MA/FA ratios and identify the most stable high-efficiency device stack.

BEST PCE

21.28%

S08

MEAN PCE

19.79%

8 samples

STABILITY

95%

best retained

MEAN VOC

1.09 V

cohort

OPEN ISSUES

3

quality review

PROJECT

PRJ-2026-014

PIPELINE

chose

EVIDENCE

28 files | 18 measurements | 7 findings

Traceable preparation and device architecture

3 experiments

Versioned preparation, mapped JV data and deterministic comparison.

SOLUTION REVIEW | SOL-B04

DMF 80%

DMSO 20%

TARGET	FA0.90MA0.10PbI3
VOLUME	2.00 mL
MOLARITY	1.25 M
STATUS	Reviewed

STACK REVIEW | STK-003/V2

Glass / ITO	1.1 mm
SnO2	30 nm
FA0.90MA0.10PbI3	620 nm
Spiro-OMeTAD	180 nm
Au	80 nm

n-i-p reference architecture

EXPERIMENT COVERAGE

EXP-041

S01 | S02 | S03

CHOSE Standard v2 | 100C | 6 measurements

EXP-052

S04 | S05

CHOSE Standard v2 | 105C | 4 measurements

EXP-067

S06 | S07 | S08

CHOSE Standard v2 | 100 unit missing | 24 measurements

DATA-QUALITY REVIEW

1 ERROR

Device count mismatch

2 WARNINGS

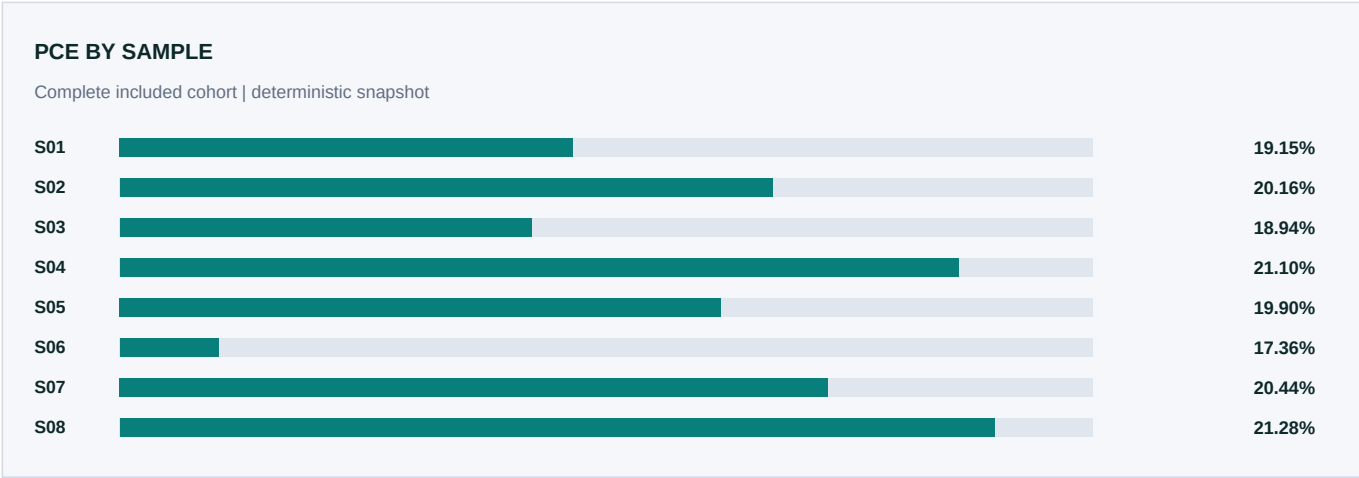
Annealing unit and solution provenance

BOUNDARY

No causal claim from incomplete metadata

Device performance and source-aligned data

8 samples | 9 fields



SAMPLE	FORMULATION	BATCH	VOC	JSC	FF	PCE	STAB.	HYST.
S01	FA0.85MA0.15	B01	1.08	22.7	78.1	19.15	89%	3.2%
S02	FA0.85MA0.15	B01	1.10	23.2	79.0	20.16	91%	2.8%
S03	FA0.80MA0.20	B02	1.07	22.9	77.3	18.94	85%	4.1%
S04	FA0.90MA0.10	B03	1.12	23.5	80.2	21.10	94%	2.1%
S05	FA0.90MA0.10	B03	1.09	23.0	79.4	19.90	92%	2.5%
S06	FA0.75MA0.25	B04	1.05	21.8	75.8	17.36	78%	5.9%
S07	FA0.85MA0.15	B05	1.11	23.1	79.7	20.44	90%	2.7%
S08	FA0.90MA0.10	B06	1.13	23.4	80.5	21.28	95%	1.9%

RESEARCHER INTERPRETATION

S08 remains the current performance leader.

04 | EVIDENCE-LINKED FINDINGS

Advisory review with explicit boundaries

5 findings

96 FA0.90MA0.10 is the strongest candidate S04 and S08 lead both PCE and stability, with the lowest hysteresis. S04, S08 7 metrics Simulated AI	ACCEPTED	91 S06 is a multi-metric outlier PCE, fill factor and stability are jointly below the robust cohort range; review fabrication notes before exclusion. S06 IQR + robust z Simulated AI	REVIEW
88 Batch effect is smaller than formulation effect Within-batch variation is limited compared with the shift between formulations. Grouped comparison Simulated AI	ACCEPTED	84 Stability and hysteresis are inversely associated Lower hysteresis appears in the most stable devices; the sample count does not support a causal claim. Spearman preview Simulated AI	REVIEW
78 Two metadata gaps limit reproducibility Humidity during coating and elapsed time before annealing are missing for B04 and B05. Process records Simulated AI	ACTION		

05 | RESEARCHER DECISION

Interpretation and scientific boundaries

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DISCUSSION The current cohort supports validation but not causal inference.	CONCLUSIONS S04 and S08 proceed to validation; S06 remains under review.	LIMITATIONS Small demonstration cohort; no causal inference.
Additional notes The author requested one additional validation run.		

07 | PROVENANCE & APPROVAL

Evidence classes and final state

RAW Local source-aligned measurements	CALCULATED Deterministic KPI and comparisons
RESEARCHER Objectives, interpretation and approval	AI Simulated advisory findings requiring review
SOURCE APPENDIX batch_B03_forward.csv process_metadata.yaml SOL-B04 STK-003/v2 28 project files 18 measurements 6 linked knowledge items	
APPROVAL STATE Pending researcher signature	